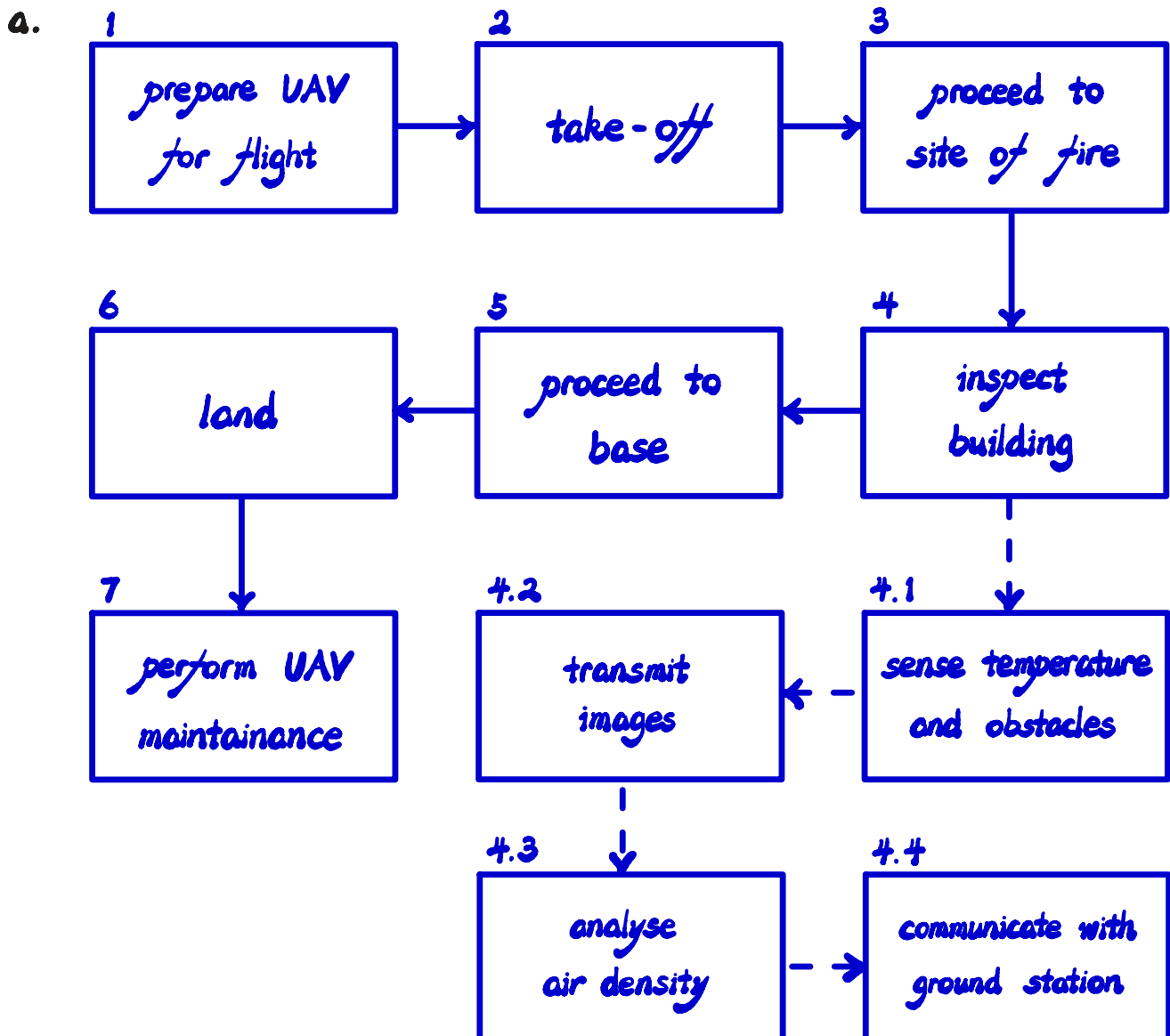


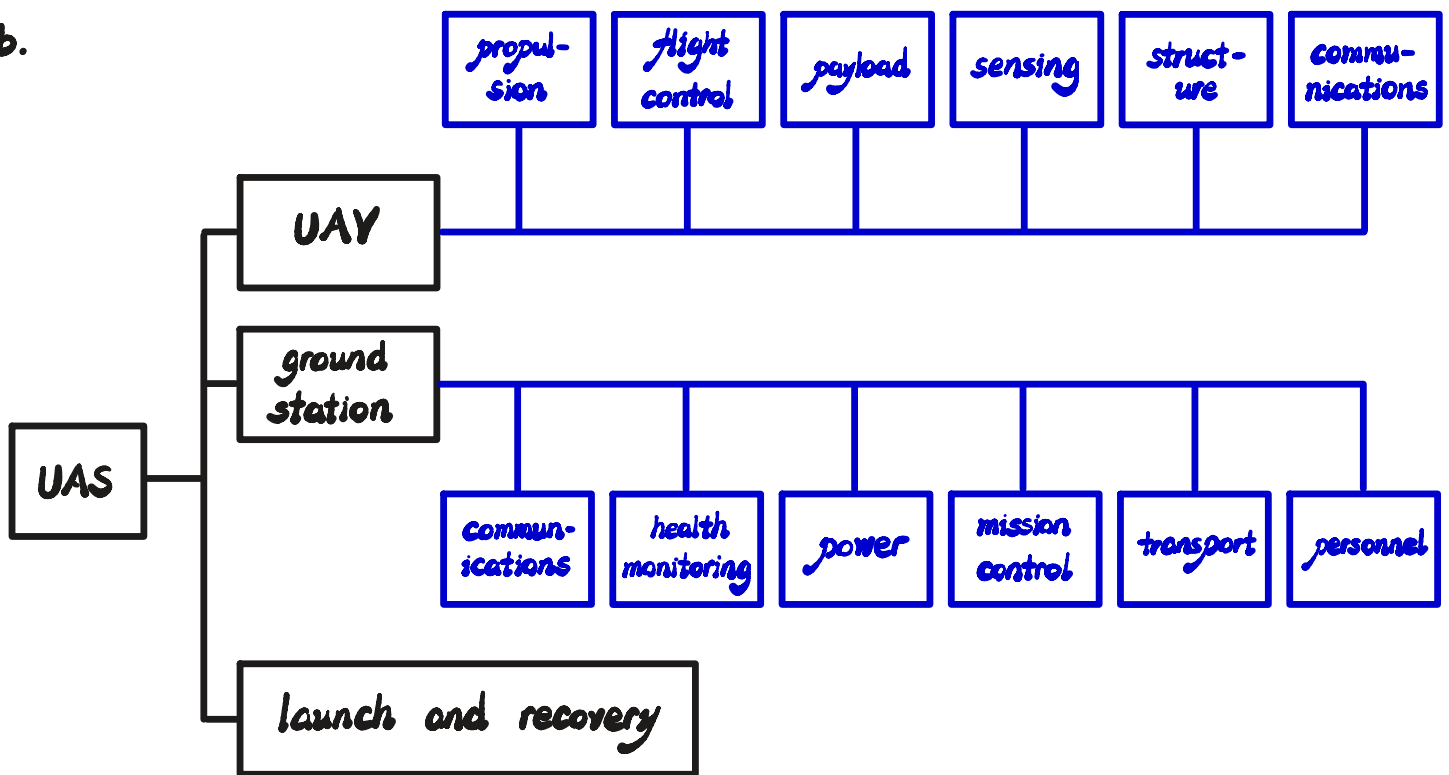
AERO70036 Systems Engineering for UAVs: Tutorial

2026

1. **Functional block diagram.** Consider a UAV, including ground station, to be used by the fire brigade to inspect buildings after fires or explosions:
 - a. Draw a functional block diagram of its mission.
 - b. Give a breakdown of the UAV system into sub-systems. Justify your choice based on the functions above.



b.



2. N2 chart. Consider an “ironing system” with, at least, the following elements: steam iron, ironing board, electrical network, and user.

- What are the “user needs” for this system?
- Is the system complete?
- Name the primary functions of each these elements and any additional elements to complete your system.
- Draw a N2 chart to identify the interfaces between elements. Describe each of the interfaces in one or two words.

a. smoothen clothing items with different materials

b. no, its missing (i) water supply and (ii) item to be ironed

c. iron : (i) exert pressure on item ; (ii) heat item ; (iii) humidity item ; (iv) HMI

ironing board : (i) support item ; (ii) support iron

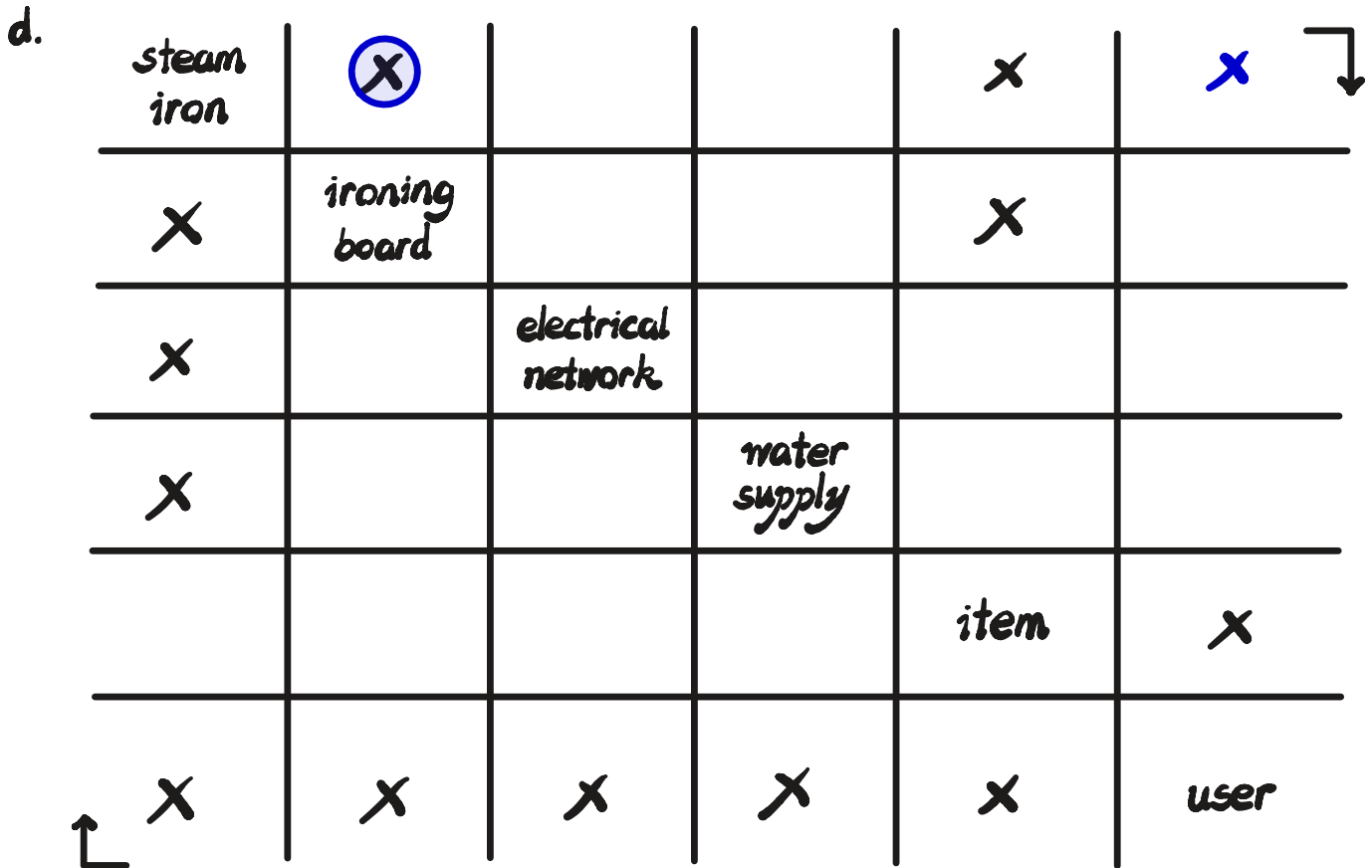
electrical network : (i) provide energy to steam iron

water supply : (i) provide water for steam iron

item : (i) being ironed ; (ii) provide clothing to iron

user : (i) control iron ; (ii) control item to iron ; (iii) set up system ;

(iv) provide water to iron



3. **Risk assessment.** You are asked to perform a technical risk assessment of the flight phase of the UAV via a risk map:

- Which two factors determine the risk in a risk map? How do you evaluate risk?
- Draw a risk map of the following essential UAV functions: communications, propulsion, sensing, flight control, structural integrity, and response to environment.
- Which two typical risk mitigation approaches might be taken? Indicate how the highest risks in the map can be mitigated.

a. $\text{risk} = \text{likelihood} \times \text{impact}$

b. see solution

c. i) change the likelihood : use more robust fly controller

ii) change the impact : adding failsafe layer on receiver

4. **System reliability.** The reliabilities of three components denoted A, B and C are 0.98, 0.85 and 0.88, respectively. Determine the reliability of a system consisting of the three components connected:
- In series,
 - In parallel.
 - In the system configuration depicted in figure 1

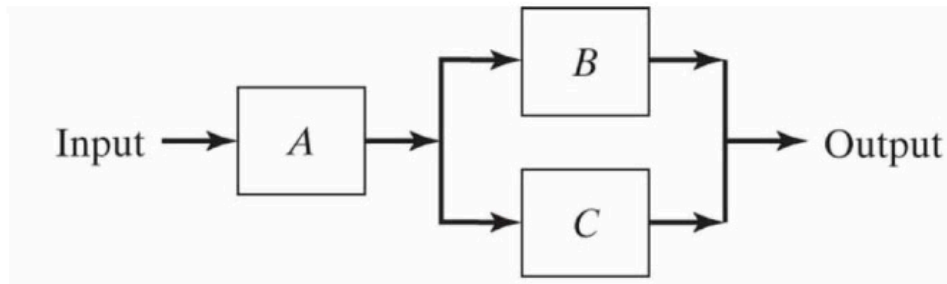


Figure 1: Series-parallel network.

$$0.98 \cdot 0.85 \cdot 0.88 = 0.73$$

$$a. R = e^{-0.98t} \cdot e^{-0.85t} \cdot e^{-0.88t} = e^{-2.71t}$$

$$1 - (1 - 0.98)(1 - 0.85)(1 - 0.88) = 0.9996$$

$$b. R = (1 - e^{-0.98t})(1 - e^{-0.85t})(1 - e^{-0.88t})$$

$$= (1 - e^{-0.98t} - e^{-0.85t} + e^{-1.83t})(1 - e^{-0.88t})$$

$$= 1 - e^{-0.98t} - e^{-0.85t} + e^{-1.83t} + e^{-1.86t} + e^{-1.73t} - e^{-2.71t}$$

$$0.98 \cdot [1 - (1 - 0.85)(1 - 0.88)] = 0.96$$

$$c. R = e^{-0.98t} (1 - e^{-0.85t})(1 - e^{-0.88t})$$

$$= e^{-0.98t} (1 - e^{-0.85t} - e^{-0.88t} + e^{-1.73t})$$

$$= e^{-0.98t} - e^{-1.83t} - e^{-1.86t} + e^{-2.71t}$$

5. **System reliability.** Calculate the reliability of a system consisting of two units, one operating and the other in standby, operating for 200 hours. The failure rate of each unit is 0.003 failures per hour and the failure-sensing switch is 100% reliable.

$$\begin{aligned}
 R &= (1 + \lambda t) e^{-\lambda t} \\
 &= (1 + 0.003) e^{-0.003} \\
 &= 0.999996
 \end{aligned}$$

6. **House of Quality.** A HoQ is to be employed for the requirement analysis for improving the design of a rucksack. Two competitor designs have the characteristics given in table 1. A market research provided the evaluation of the two designs and your design is given in table 2 (1=worst to 5=best).

- What is the primary function of the HoQ?
- What are the main component matrices of the HoQ and describe their purpose?
- Construct the HoQ for the analysis of requirements for the rucksack. Discuss how you can use this information to set suitable targets for the characteristics of the rucksack.

Competitor	A	B
❶ Number of compartments	2	3
❷ Weight of rucksack (kg)	1.5	2.0
❸ Max. capacity (litres)	25	35
❹ Cost (pounds)	20	30
❺ Durability of colour	High	Medium

Table 1: Rucksack characteristics.

Customer view	Importance?	A	B	Yours
Durable	Very high	4	3	5
Lightweight	High	2	3	3
Roomy	Very high	3	4	3
Attractive	High	4	3	2
Low cost	Low	2	3	4

Table 2: Market research.

- implement the quality functional deployment (QFD), i.e. translate customer needs to system requirement to technical solutions, also, HoQ identify and

prioritise the technical performance measures (TPMs).

b. see solution

c. see solution

Short conceptual questions

1. What is verification? What is validation? Name and briefly describe three methods for system verification.

2. Name four human-system interface issues that affect the effectiveness of a UAS.
3. Name two civilian UAS missions. What are their main payloads? What class of UAVs are suitable for these missions?
4. What is meant by functional analysis in systems engineering? When in the lifecycle is it accomplished? What purpose does it serve? Identify some of the benefits derived. Can a functional analysis be accomplished for any system?
5. What is the meaning of "designing for the full lifecycle"? What are requirements? List the desirable characteristics of a description of requirements.
6. What is it meant by human factors? Why is it important to consider human factors in design? When in the lifecycle of the system should they be considered and why?

1. *verification* : is the system right ? i.e. are requirements meet ?

i) inspection ; ii) analysis ; iii) demonstration ; iv) test ; v) certification .

validation : is it right system ? i.e. is system fit ^{for} purpose ?

2. i) *workspace* : control station is dirty and noisy ;

ii) *personnel* : operator lack of training and stressful ;

iii) *task* : poor design of the task of UAV ;

iv) *organisational* : poor leadership .

3. *film, photography* - camera - mUAVs, mainly quadcopters ;

weather monitoring - sensor to monitor weather - HALE .

4. • *determine function of system and allocate function to components*
 • *mainly in soft (conceptual and definition) phases.*
 • *early stage of conceptual design*

- basis of development of detailed design requirement
- yes

5. • perform design throughout CD*BTO (R²)

- essential condition and properties that system and its elements must satisfied

• SMART :

- specific ;
- measurable ;
- achievable / attainable ;
- ~~reliable~~ ; relevant ;
- timely.

6. • concern with the integration of human issues within a system.

• humans :

- are major performance driver ;
- have major impact on system safety ;
- are major cost driver through the lifecycle.

• all

Additional questions

1. Identify in a jet aero-engine Hillary Sillitto's "Ten things to understand about systems"¹:
 - (a) A system exists within a wider "context" or environment.
 - (b) A system is made up of parts that interact with each other and with the wider environment.
 - (c) A system has structure, function, performance, behaviour and a lifecycle.
 - (d) A system has system-level properties ("emergent properties") that are properties of the whole system not attributable to individual parts.
 - (e) A system both changes its environment and adapts to its environment when it is deployed.
 - (f) Systems contain multiple feedback loops with variable time-constants.
 - (g) A system may be part of one or several wider "containing systems."
 - (h) A system may have one of three basic types of relationship with its environment: distinct, close-coupled, or fluid and dynamic.
 - (i) A system may offer "affordances" for interaction.
 - (j) Types of system include technical, biological, social, ecological, environmental and any combination of these.

see solutions

¹"Architecting Systems", Hilary Sillitto, College Publications, 2014.

2. Identify the systems involved in the example of system interactions concerning heat transfer in Figure 2. Use an N2 chart to identify the (directional) interactions between systems.

¹"Architecting Systems", Hilary Sillitto, College Publications, 2014.

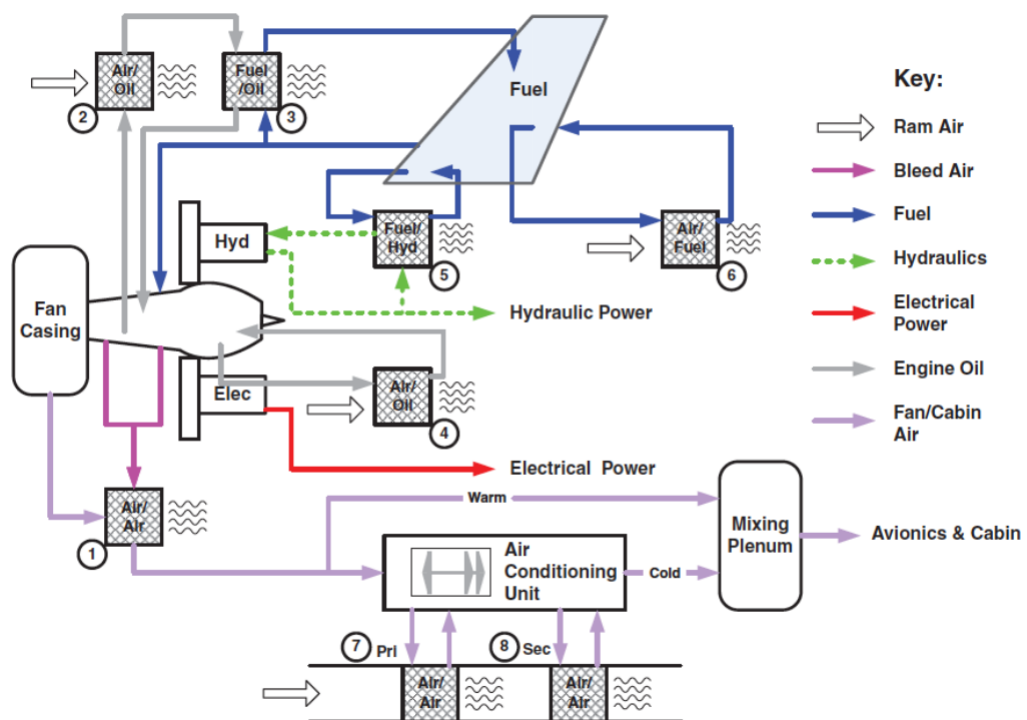


Figure 2: An example of system interactions (Fig. 2.7 M&S).

see solutions